Mongolia International Mathematics Competition 2026

Ulaanbaatar City, 22nd to 27th July 2026

Invitational World Youth Mathematics Intercity Competition

Individual Contest

Time limit: 120 minutes

Information:

- You are allowed 120 minutes for this paper, consisting of 12 questions in Section A to which only numerical answers are required, and 3 questions in Section B to which full solutions are required.
- Each question in Section A is worth 5 points. No partial credits are given. There are no penalties for incorrect answers, but you must not give more than the number of answers being asked for. For questions asking for several answers, full credit will only be given if all correct answers are found. Each question in Section B is worth 20 points. Partial credits may be awarded.
- Diagrams shown may not be drawn to scale.

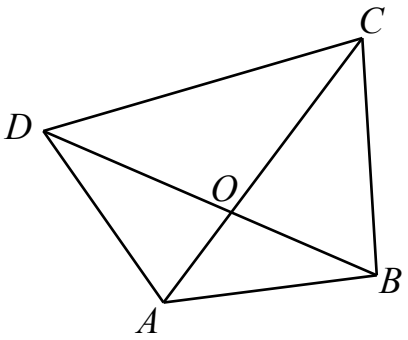
Instructions:

- Write down your serial number, name, contestant ID and team's name on the answer sheet of Section A and on **each page of Section B**.
- For Section A, write your answers in the space provided on the answer sheet.
  For Section B, write down your solutions on spaces provided after individual questions.
- You are allowed to use HB, B, and 2B pencils and ball-point pens with black or blue ink.
- You cannot use instruments such as protractors, calculators and electronic devices, smart watches.
- At the end of the contest, you must hand in the envelope containing the question sheets, the answer sheet and all scrap paper.

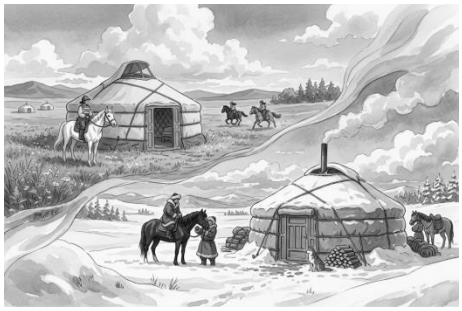
English Version

Section A.
In this section, there are 12 questions. Fill in the correct answer in the answer sheet of each question. Each correct answer is worth 5 points.

1. As shown in the diagram below, the diagonals of convex quadrilateral *ABCD* intersect at point *O*. If triangle *AOB* has an area of 144 cm^2 and triangle *COD* has an area of 169 cm^2, what is the minimum possible area, in cm^2, of quadrilateral *ABCD*?



2. A Mongolian nomadic family owns 31 identical felt rugs. They are to distribute some of the rugs (not necessarily all) between two gers: a summer ger and a winter ger, where each ger must receive at least two rugs. In how many ways can this be done?

A summer ger  A winter ger

3. If the expression
$(-x_1+x_2+x_3+x_4)^4+(x_1-x_2+x_3+x_4)^4+(x_1+x_2-x_3+x_4)^4+(x_1+x_2+x_3-x_4)^4$
is fully expanded into a polynomial and like terms are combined, the result is a polynomial in x_1, x_2, x_3 and x_4. What is the smallest coefficient that appears in this polynomial?

4. Let *a*, *b* and *c* be non-negative real numbers. Define *X*, *Y* and *Z* such that

$$X=(2a+b+c)(b+c)+c^2,$$
$$Y=(a+2b+c)(a+c)+a^2,$$
$$Z=(a+b+2c)(a+b)+b^2.$$

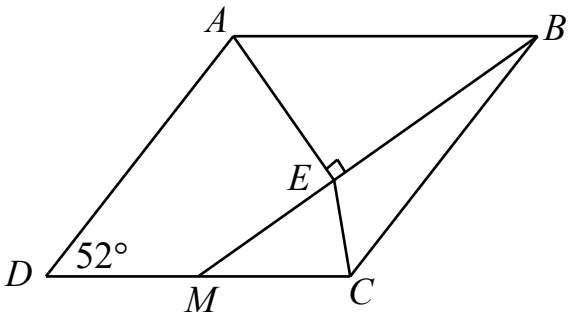
If $X+Y+Z=12$, what is the value of $\dfrac{a^3+b^3+c^3-8}{(2-a)(2-b)(2-c)}$?

5. How many ordered pairs (*a*, *b*) of positive integers are there which satisfies

$$a\times b+57=29\times\gcd(a,b)+2\times\operatorname{lcm}(a,b)\,?$$

(Note: gcd(*a*,*b*) denotes the greatest common divisor of *a* and *b*;
lcm(*a*,*b*) denotes the least common multiple of *a* and *b*.)

6. As shown in the diagram below, $ABCD$ is a rhombus with $\angle D = 52°$. Point M is the midpoint of DC and point E lies on segment BM such that $AE \perp BM$. What is the measure, in degrees, of $\angle BEC$?



7. Replacing exactly one of the six digits by another digit in the expression 435×605, the resulting product becomes N^2, where N is a positive integer. What is the numerical value of N?

8. Before the Naadam Festival, a merchant is preparing his goods to sell. He has 7 different traditional Mongolian items namely deel (tunic), airag (mare's milk), morin khuur (horsehead fiddle), shirdeg (felt rug), gutal (leather boots), malgai (hat) and boortsog (fried cookies).



The merchant can create a small package using any non-empty combination of the items, with no item included more than once. The merchant can then select two small packages that do not share any items to form a special value deal (the ordering of the packages does not matter). What is the total number of distinct special value deals that the merchant can create?

Note: The special value deal {(deel, airag, shirdeg), (morin khuur, gutal)} is considered the same as {(morin khuur, gutal), (deel, airag, shirdeg)}, but different from {(deel, airag, morin khuur), (shirdeg, gutal)}.

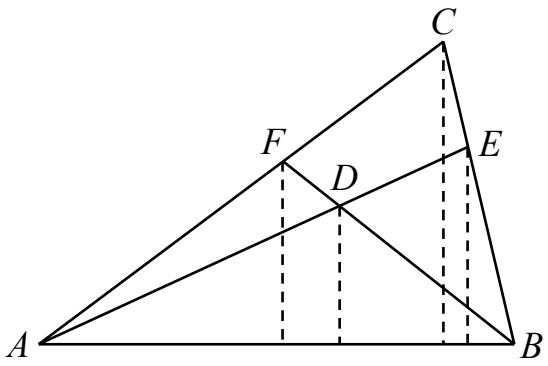
9. Let A, B and C be three-digit integers whose decimal representations together use exactly nine distinct digits. If $A < B < C$ and $A + B + C = 2026$, what is the smallest possible value of C?

10. What is the minimum value of $f(x)=\sqrt{x^2-4x+5}+\sqrt{x^2-10x+34}$ over all real numbers x?

11. An ancient Mongolian iron chest is secured by a circular lock consisting of eight empty slots arranged in a ring. To unlock the chest, an explorer must place eight sacred medallions, numbered 0, 1, 2, 3, 4, 5, 6 and 7 into the slots, such that exactly one medallion occupies each slot. According to the description on the lid, the chest only opens if the difference between every pair of adjacent medallions is a prime number (that is 2, 3, 5 or 7).

Two circular configurations are considered identical if one can be rotated to match the other. How many unique circular configurations can unlock the chest?

12. As shown in the diagram below, points E and F lie on sides BC and AC, respectively, of triangle ABC. The segments AE and BF intersect at point D. The perpendicular distances from points E, D and C to line AB are 7 cm, 5 cm and 11 cm, respectively. What is the perpendicular distance, in cm, from F to line AB?



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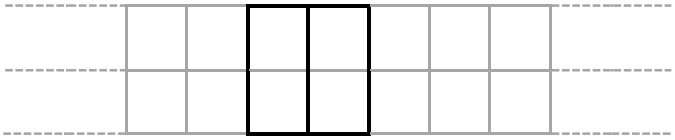
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Section B.
Answer the following 3 questions, and show your detailed solution in the space provided after each question. Each question is worth 20 points.

1. A 2×2026 rectangular board is tiled completely by using 2026 identical dominoes, each of size 2×1. Let a denote the number of such tilings in which no two vertical dominoes share a common side as shown in the diagram below. What is the remainder when a is divided by 6?



Answer: _____

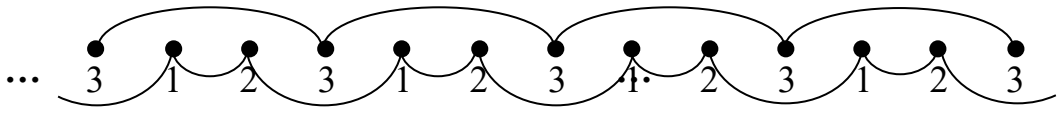
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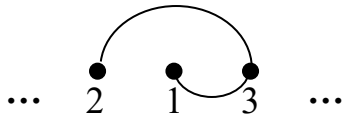
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2. On an infinite strip of adjacent squares (which can be associated with the integers), there is exactly one ball in each square. Written on these squares is a pattern of numbers that repeats periodically forever. The pattern is formed by a permutation of the numbers from 1 to n ($n \geq 3$) which we denote as $\overline{a_1 a_2 ... a_n}$.

At a given signal, each ball jumps to the right over a number of squares equal to the value written in its starting square. The pattern is called ***collision-free*** if, after the jump, there is no square in which two or more balls land. For example, the patterns $\overline{123}$, $\overline{231}$, $\overline{312}$ are ***collision-free***, as shown below:



On the other hand, the pattern $\overline{213}$ is not ***collision-free***, as shown below:



Two patterns are considered identical if they can be obtained from one another through a cyclic shift. For example, $\overline{123}$ produces the same infinite strip as $\overline{231}$ or $\overline{312}$, that is $\overline{123} = \overline{231} = \overline{312}$.

For $n = 3$, there is only one ***collision-free*** pattern.

How many ***different collision-free*** patterns exist for $n = 5$?

Explain your reasoning in complete detail. Simply listing down the correct ***collision-free*** patterns will not receive full credit.

Answer: _____ collision-free sequences

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3. As shown in the diagram below, $ABCD$ is a convex quadrilateral such that BD bisects $\angle ABC$, the circumcircle of triangle ABC intersects AD and CD at P and Q, respectively. A line drawn though D parallel to AC meets the extensions of BC and AB at R and S, respectively. Prove that P, Q, R and S are concyclic.

